

Deep Reinforcement Learning with Ring Attractor: Stable and Efficient Methods for Continuous Control Tasks

Qihao Wang Zijie Jing Zhuo Deng

Abstract: Standard deep reinforcement learning (DRL) methods for continuous control tasks, such as DDPG and SAC, treat action components independently, ignoring inherent spatial structures in many tasks. This leads to inefficient exploration and unstable training. We propose integrating the Ring Attractor (RA), a neuroscience-inspired recurrent network with ring topology into DRL frameworks to explicitly encode spatial continuity. Our RA-driven action selection mechanism leverages competitive neurodynamics to refine Actor outputs guided by Critic network evaluations, providing three key benefits: (1) manifold-constrained exploration that compresses the search space, (2) Winner-Take-All dynamics for decisive action selection in ambiguous scenarios, (3) neurodynamic filtering for gradient stabilization. Experimental results on Pendulum-v1 demonstrate that RA-augmented algorithms (DDPG-RA, SAC-RA) achieve **46-52%** faster convergence compared to baselines, validating the effectiveness of biologically-inspired spatial representations for continuous control tasks. Our codes are available at: <https://github.com/Dawnlast/DRL-RA>.

Keywords: reinforcement learning, ring attractor, continuous control, neurodynamics, spatial structure

1 Introduction

Reinforcement learning (RL)[1] offers a mathematical framework for optimizing decision-making strategies through environmental interaction, typically formalized as a Markov Decision Process (MDP)[2]. The primary objective of the agent is to learn an optimal policy π that maximizes the expected cumulative reward G : $\pi^* = \operatorname{argmax}_{\pi} \mathbb{E}_{\tau \sim \pi}(G) = \operatorname{argmax}_{\pi} \mathbb{E}_{\tau \sim \pi} [\sum_{t=0}^{\infty} \gamma^t r_t]$.

For high-dimensional continuous action spaces in domains like robotics and autonomous driving, Deep Deterministic Policy Gradient (DDPG)[3] and Soft Actor-Critic (SAC)[4] represent the state-of-the-art solutions. However, these methods treat action components independently, failing to capture inherent spatial structures. In navigation tasks, for instance, adjacent directions possess spatial continuity that fully connected layers cannot explicitly encode, leading to suboptimal learning efficiency.

To address this limitation, we incorporate the RA model[5] from neuroscience. The RA is a recurrent neural network-like model originally identified in brain navigation circuits, encoding head direction and spatial position. The RA architecture comprises excitatory neurons arranged in a ring with strong local connections, and a central inhibitory neuron providing global feedback to prevent over-activation. Through competitive dynamics, the RA can help the agent quickly converge to one of the attractor states in decision-making.[5] According to this characteristic, we combine RA with DDPG and SAC to enhance the stability and convergence efficiency of the algorithms in continuous control tasks, as is shown in Figure 1.

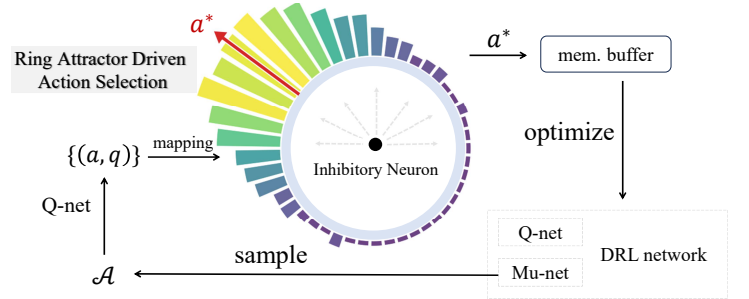


Figure 1: DRL combined with RA for continuous control tasks.

2 Methodology: Ring Attractor Based Action Selection

We propose a neurodynamics-based action selection mechanism that integrates the RA model with DRL frameworks. Given the state s_t , we construct an action set $\mathcal{A} = \{a_0, a_1, \dots, a_K\}$ comprising the Actor output $a_0 = \mu_{\theta}(s_t)$ and exploratory perturbations $a \sim \mathcal{N}(a_0, \sigma)$. The Critic evaluates these actions, yielding Q-values $q_k = Q(s_t, a_k)$ for $k \in \{0, 1, \dots, K\}$. These Q-values are then converted into external input currents x_n for the RA network. For the n -th neuron with preferred action angle θ_n , the input current is:

$$x_n = \sum_{k=0}^K (\tilde{q}_k \cdot \alpha + \beta) \exp \left(-\frac{(\theta_n - \psi(a_k))^2}{2\sigma_{\text{ext}}^2} \right) \quad (1)$$

where $\tilde{q}_k$ is the normalized Q-value, $\psi(\cdot)$ maps actions onto ring topology $[0, 2\pi)$, σ_{ext} determines input spread, and α, β are fixed scaling parameters. This ensures high-Q actions deliver stronger excitatory input.

The network evolves according to neurodynamic equations (2). Recurrent excitatory connections $w^{(E \rightarrow E)}$ amplify activity v_n in high-Q regions, while global inhibitory feedback $w^{(I \rightarrow E_u)}$ suppresses suboptimal areas.

$$\frac{dv_n}{dt} = -v_n + \frac{1}{\tau} \cdot \text{ReLU} \left(x_n + \sum_{m=1}^N w_{m,n}^{(E \rightarrow E)} v_m + w^{(I \rightarrow E_u)} u \right), \quad \frac{du}{dt} = -u + \frac{1}{\tau} \cdot \text{ReLU} \left(\sum_{n=1}^N w_n^{(E_n \rightarrow I)} v_n + u \right) \quad (2)$$

The executed action a^* is decoded via population vector decoder: $a^* = \psi^{-1} \left(\arg \sum_{n=1}^N v_n^{(T)} e^{i\theta_n} \right)$, where $v_n^{(T)}$ is the activity at time T , and $\arg(\cdot)$ extracts the phase angle to $[0, 2\pi)$. This mechanism leverages the RA’s manifold structure to filter exploration noise, converging to optimal actions.

3 Mechanism Analysis & Experimental Verification

In conventional DRL, policy networks are typically implemented using MLPs, which are unconstrained function approximators prone to high-frequency noise. In contrast, RA introduces crucial inductive biases through three key mechanisms.

First, topological structure: The RA’s recurrent dynamics constrain actions to a low-dimensional ring manifold. Regardless of the Actor’s initial output a_0 , if it falls within the attractor’s basin $\mathcal{B}$, the neurodynamics robustly guide it onto the manifold surface, filtering discontinuous actions and compressing the search space to a structured subspace.

Second, Winner-Take-All dynamics: Agents often face multiple seemingly optimal choices. For example, when encountering an obstacle, both turning left ($a = -1$) and right ($a = +1$) may be viable, while continuing straight ($a = 0$) causes collision. Traditional MLPs linearly interpolate: If $Q(s, -1) \approx Q(s, +1)$, the Actor may output $a \approx 0$. The RA’s nonlinear lateral inhibition resolves this: The global inhibitory neuron u enforces competitive dynamics, where inhibitory feedback $w^{(I \rightarrow E)}u$ suppresses weakly activated neurons while amplifying dominant signals, forcing the system to commit to a single attractor state.

Third, signal denoising: The RA acts as a dynamic low-pass filter, smoothing high-frequency fluctuations through temporal integration and spatial smoothing. This creates a synergistic feedback loop: (1) RA provides stable actions a^* to the Critic; (2) reduced noise enables accurate value estimation $Q(s, a)$; (3) precise Q-values furnish low-variance policy gradients. This variance reduction is particularly amplified in high-dimensional action spaces where exploration noise dominates the learning performance.

We hypothesize that the RA exhibits adaptive self-regulation during training. As is shown in Figure 2(a), early phases feature lower neural activity allowing broader exploration, while late phases show high-intensity peaked activity consolidating the optimal policy. Results in Figure 2(b) demonstrate that the RA enhances value estimation stability, reduce ineffective exploration, and improve exploitation efficiency, thereby significantly accelerating convergence.

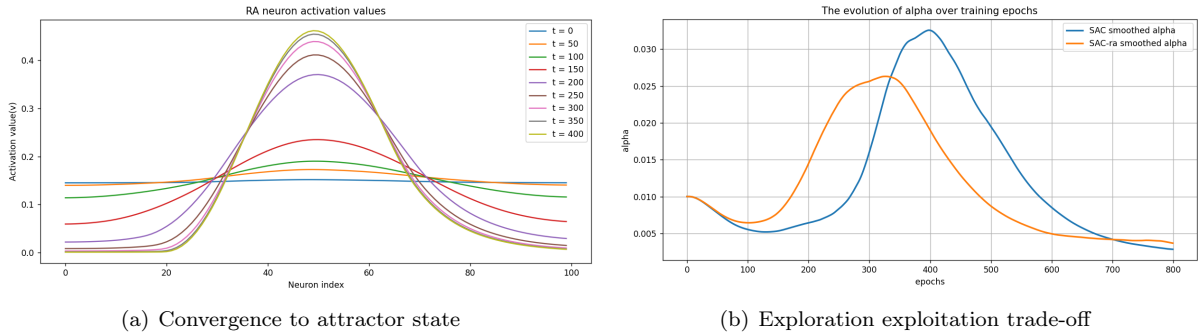


Figure 2: Ring Attractor neurodynamics and autonomous adaptation in SAC-RA during training. (a) Temporal evolution of excitatory neuron activity $v_n^{(t)}$ across neuron indices. (b) SAC-RA exhibits consistently lower α values throughout training, with the peak occurring earlier compared to vanilla SAC. Smaller α indicates reduced exploration and enhanced exploitation.

4 Results & Discussion

We evaluate our RA-augmented algorithms on the Pendulum-v1 benchmark from OpenAI Gymnasium[6].

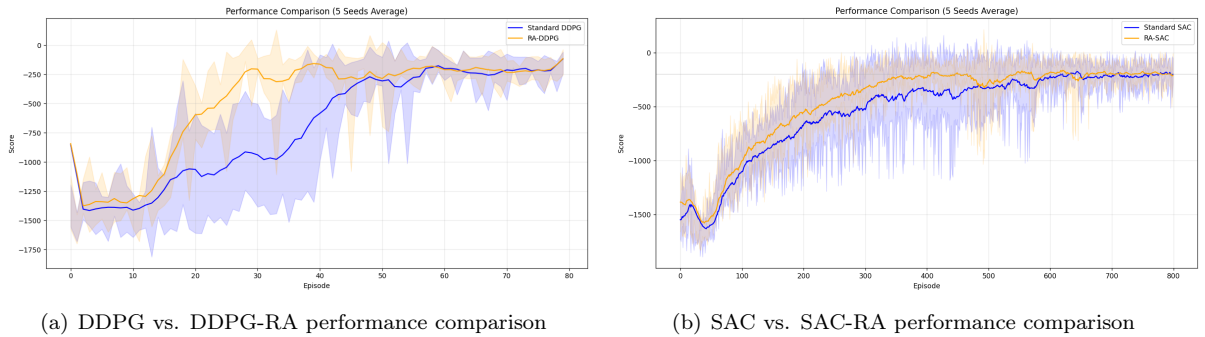


Figure 3: Learning curves on Pendulum-v1. RA-augmented algorithms (DDPG-RA, SAC-RA) demonstrate faster convergence compared to baselines (DDPG, SAC).

As shown in Figure 3, RA-augmented algorithms consistently outperform baselines across all metrics. DDPG-RA reaches target performance (-250 score) **51.8%** faster (epoch 27 vs. 56). SAC-RA reaches target performance (-200 score) **46.2%** faster (epoch 350 vs. 650). These results validate the effectiveness of RA integration in enhancing learning efficiency and stability in continuous control tasks.

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